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COMPUTING FOUNDATION

GitOps with ArgoCD

Introduction - Who am i?

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GOLDEN Kubestronaut

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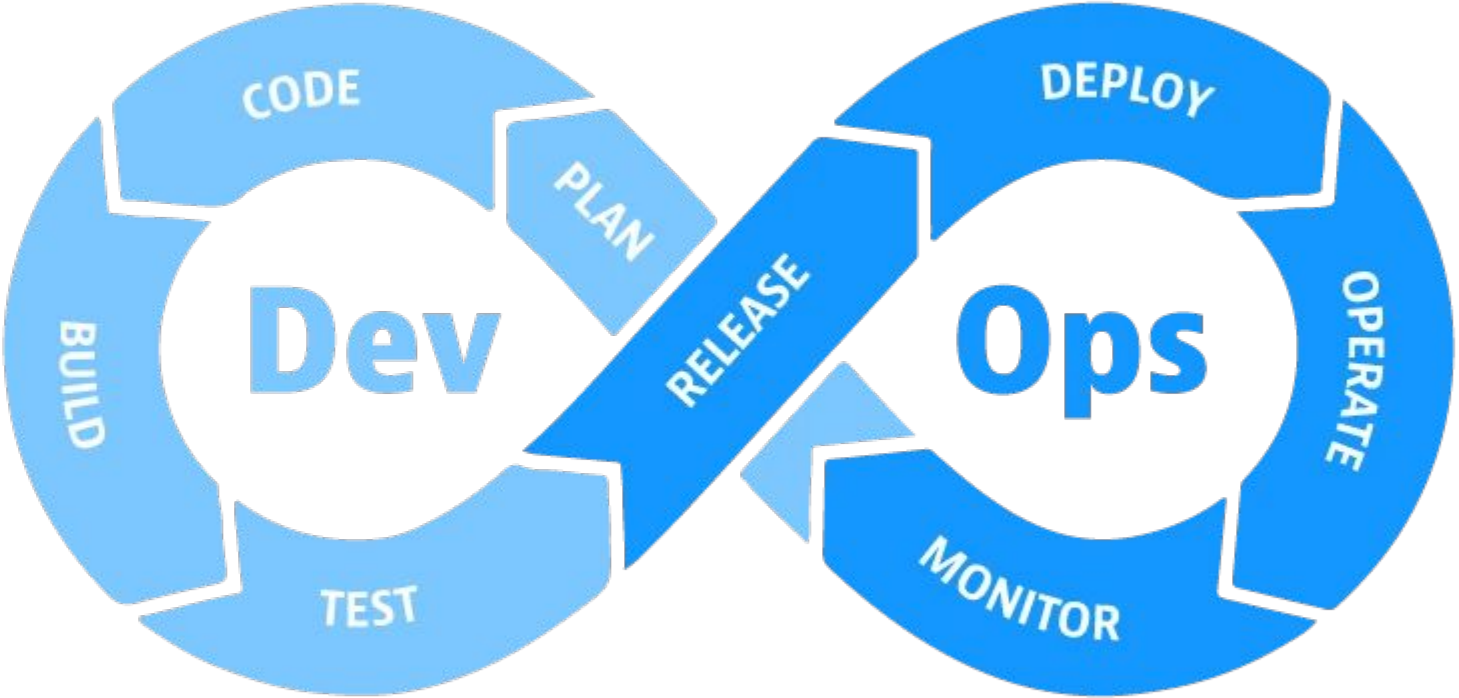
CODE
PLAN
BUILD
Dev
TEST
RELEASE
DEPLOY
Ops
OPERATE
MONITOR

What is GitOps?

GitOps is a **DevOps** practice that uses **Git** as the “**single source of truth"** for infrastructure and application deployments. When changes are made to the Git history, **GitOps** operators automatically detect and apply these changes to the **target environment**.

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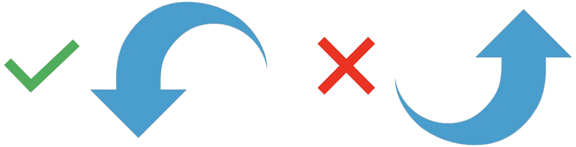
GitOps Principles

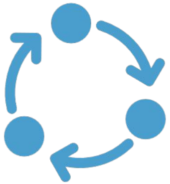
Desired State must be...

Declarative



Pulled Automatically

Versioned and Immutable



Continuously Reconciled

Imperative

“Where would you like to go today?”

- Drive out to Charoen Nakhon Road.
- Head toward Krung Thon Buri Road and go across Saphan Taksin Bridge.
- Drive straight along Rama IV Road.
- Turn right into Asok Montri Road (Sukhumvit Soi 21).

Declarative

"Where would you like to go today?"

"I want to go to this location... 13.737234137713237, 100.56192005349098"

(Soi Cow boy)

Imperative Vs Declarative

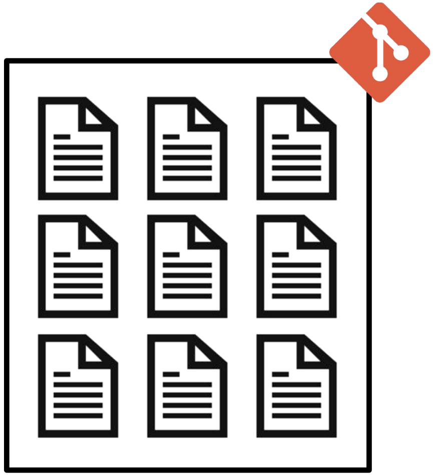
```
pipeline {
    agent any

    stages {
        stage('Deploy to Kubernetes') {
            steps {
                sh 'kubectl create deployment my-app --image=my-app:v1.2.3 || true'
                sh 'kubectl scale deployment my-app --replicas=3'
                sh 'kubectl expose deployment my-app --type=ClusterIP --port=80 || true'
            }
        }
    }
}
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 3
  template:
    spec:
      containers:
      - name: my-app
        image: my-app:v1.2.3

---

apiVersion: v1
kind: Service
metadata:
  name: my-app
spec:
  ports:
  - port: 80
  selector:
    app: my-app
  type: ClusterIP
```

Declarative

"A system managed by GitOps must have its desired state expressed ***declaratively****."*

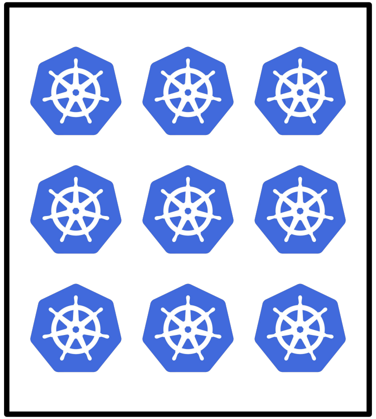
Desired state and Actual state



Desired state

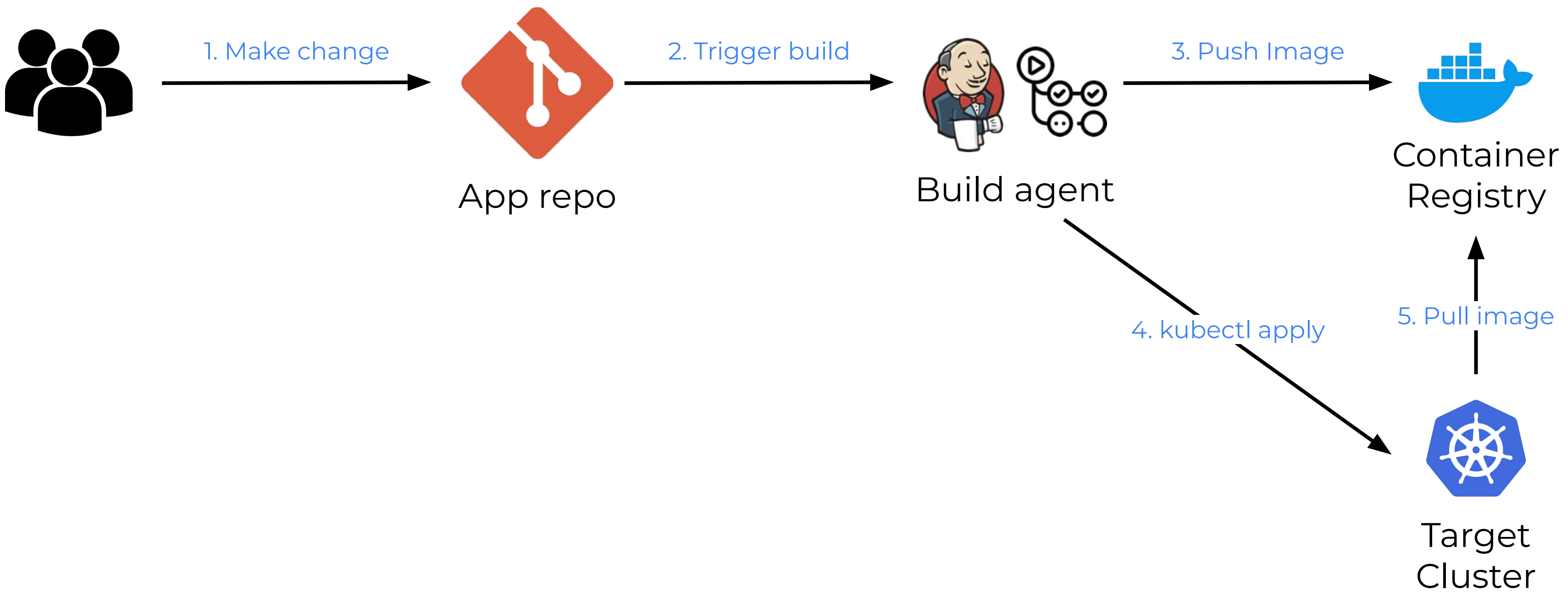
Versioned and Immutable

*"Desired state is stored in a way that enforces **immutability**, **versioning** and retains a complete version history."*

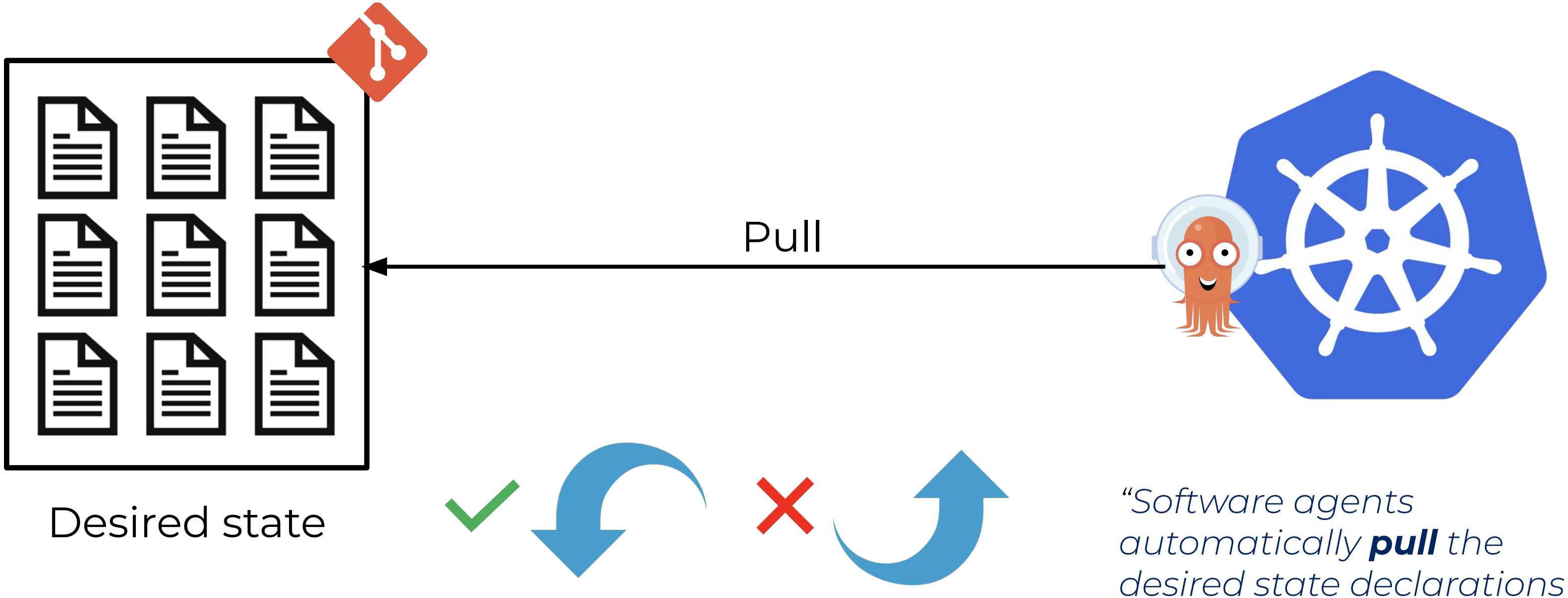


Actual state

Push-based Deployment



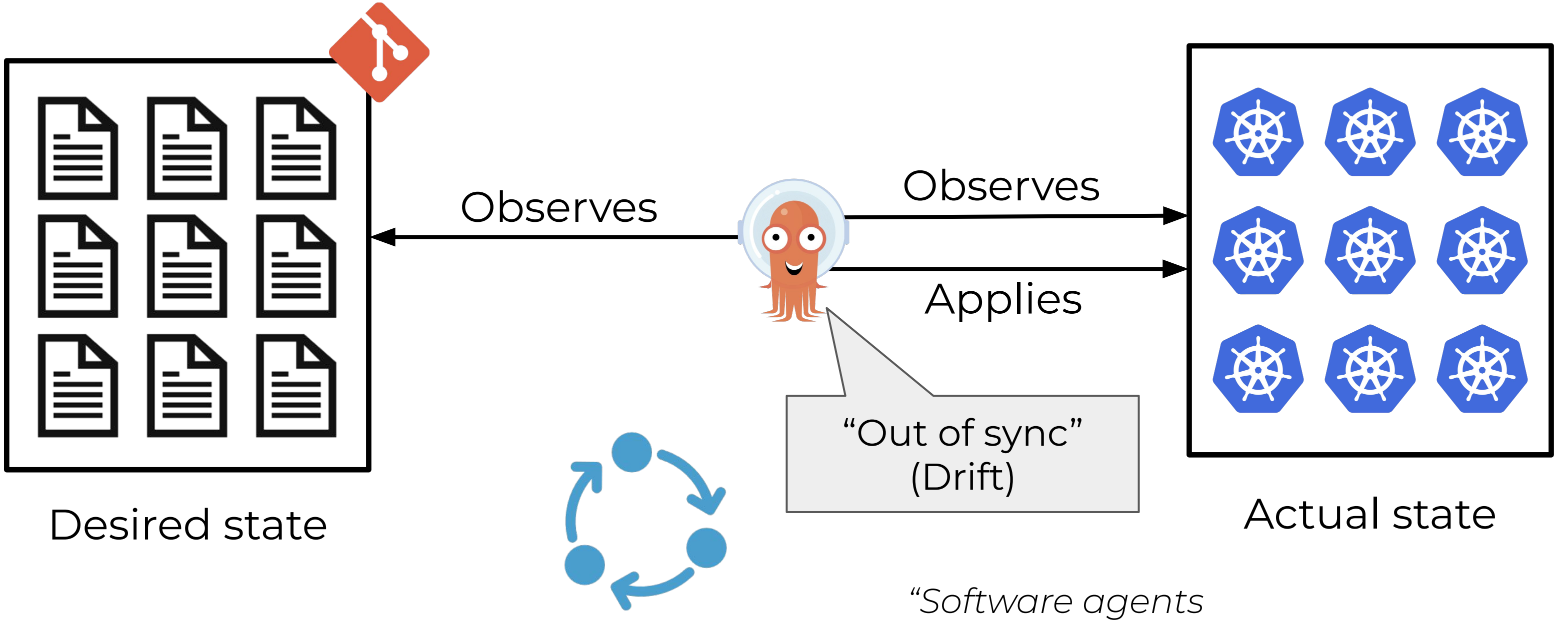
Pull-based Deployment



Pulled Automatically

*"Software agents automatically **pull** the desired state declarations from the source."*

Continuously Reconciled



Continuously Reconciled

"Software agents continuously observe actual system state and attempt to apply the desired state."

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Why/How GitOps?

Pro

- Vendor-neutral and lightweight
- Developers are already familiar with **Git**
- Eliminate configuration drift
- Clearly track history in **Git**
- Reproducible, immutable and safer deployments

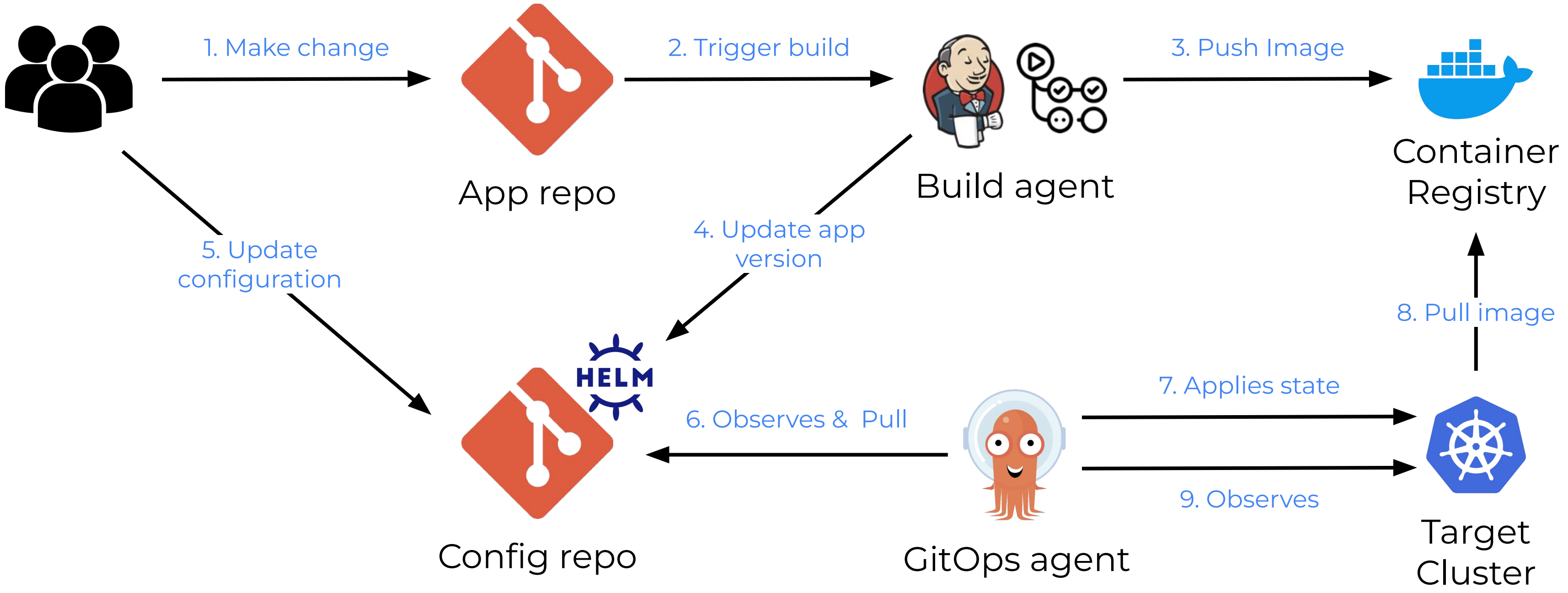
Con

- Secret management is quite challenge

AWS Secrets Manager

- Repo Sprawl
- GitOps covers only a subset of the software lifecycle (CD)

GitOps practice



	Traditional CI/CD Tools	GitOps Operators
Declarative	can deploy declarative manifests. But it is normally used for performing steps.	Designed for declarative application deployments, Infrastructure (E.g., Kubernetes manifests)
Versioned and Immutable	Even though the pipelines or state are normally stored in Git, But it doesn't enforce immutability.	Everything must be stored in Git including all configurations and the application version.
Pulled Automatically	follows a Push-based model, where changes are pushed to the cluster after a pipeline runs	GitOps operators operate on Pull-based deployment
Continuously Reconciled	No reconciliation loop. It performs the deployment but doesn't monitor the actual state	Built-in reconciliation loop. It performs the deployment and continuously remediates the drifts between the target system and the desired state

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ArgoCD

Argo Workflow

Argo CD

Argo Rollout

Argo Event

- 1 of 4 siblings under Argo project
- Graduated (6 Dec 2022)
- Supports Helm, Kustomize

<< Check this out !

Who else uses Argo CD?

[argo-cd/USERS.md at master · argoproj/argo-cd · GitHub]()

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: cncf-demo-app
  namespace: argocd
spec:
  project: default

  source:
    repoURL: https://github.com/jaruwat-panturat/k8s-lab.git
    targetRevision: main
    path: cncfDemo

  destination:
    server: https://kubernetes.default.svc
    namespace: cncf-demo

  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

Where do you store the desired state?

Where do you want to deploy the application (actual state)?

How do you want to apply changes?

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GitOps Learning Resources

OpenGitOps

https://opengitops.dev

Introduction to GitOps (LFS169)

https://training.linuxfoundation.org/training/introduction-to-gitops-lfs169/

Linux Foundations - Free courses

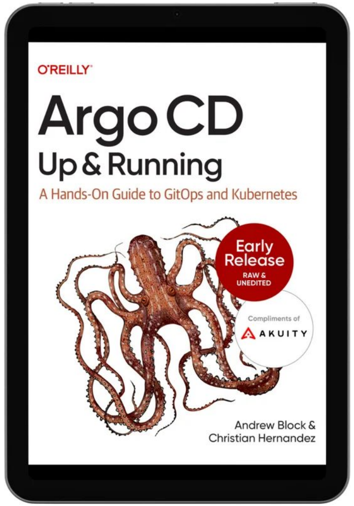
https://training.linuxfoundation.org/resources/?_sft_content_type=free-course

Killer Coda - Scenarios for the Argo Project (Free)

https://killercoda.com/argo

GitOps fundamentals course from Codefresh (49.95$)

https://learning.codefresh.io/course/gitops-fundamentals



Free book - Argo CD: Up and Running

https://landing.akuity.io/resources/argo-cd-up-and-running

[Certified GitOps Associate (CGOA)]

[Certified Argo Project Associate (CAPA)]